Terminal

For each exercise, use Terminal only! Do not rely on the Jupyter GUI.

1. Use terminal commands to navigate through directories. Start in your home directory, navigate to a subdirectory named "work," list the contents of the directory, and then return to your home directory.

2. Create a new directory named "Exercise2," navigate into the directory, and create three empty text files named "file1.txt," "file2.txt," and "file3.txt." List the contents of the directory to verify the creation of the files.

3. Create a new C++ source file named "program.cpp" in the "Exercise2" directory. Write a simple "Hello, World!" program in the file, including the necessary main function. Use the g++ compiler to compile the program into an executable named "program." Run the executable to see the output.

4. Create a new C++ source file named "program.cpp" in the "Exercise2" directory. Write a simple program that prompts the user to enter their name and greets them with a personalized message. Use the necessary C++ syntax, including the main function. Use the g++ compiler to compile the program into an executable named "program." Run the executable and provide your name to see the personalized greeting.

5. Navigate to the "Exercise2" directory. Create a new file named "textfile.txt" and write some content to it using a text

editor. Use the "ls -l" command to display the file's permissions and verify that you have read and write permissions for the file. Change the permissions of the file to only allow read access for others. Verify the updated permissions using the "ls -l" command again.

6. Navigate to the parent directory of "Exercise2." Use the "ls" command to list the contents of the directory. Sort the contents in reverse order using the "ls -r" command. Sort the contents by modification time using the "ls -lt" command. Observe the different listings and note the changes in order.

7. Navigate to the "Exercise2" directory. Delete the file named "textfile.txt" using the "rm" command. Verify that the file has been removed by listing the contents of the directory. Delete the "Exercise2" directory and its contents using the "rm -r" command. Confirm the removal by listing the contents of the parent directory.

8. Navigate to a directory that contains multiple files and subdirectories. Use the "find" command to search for a specific file within the directory and its subdirectories. Use appropriate options to control the search depth, file type, or other criteria. Verify that the file is found and note its location.